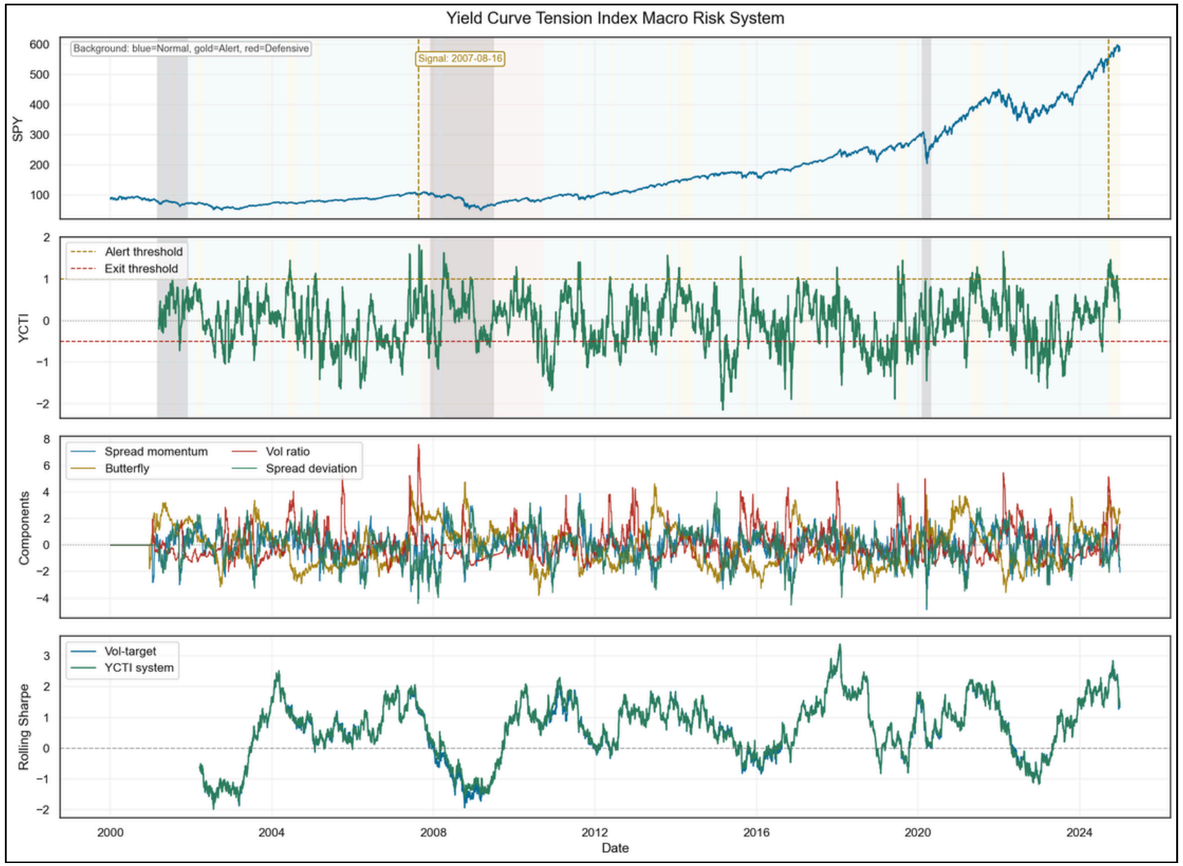
Yield Curve Tension Index: Quantifying Macro Stress for Risk-Aware Equities Trading

The yield curve has historically warned of recessions when it inverts. I tested whether that signal is strong enough to use in equity markets, and found the real danger is not the inversion itself but the moment it un-inverts.



YCTI overlay on SPY 2000-2024.
Signal fired August 16, 2007 — 347 days before the GFC trough in the 2008 recession.

A four-component Yield Curve Tension Index (YCTI) that detects this un-inversion pattern and conditions equity risk exposure accordingly.

Key Findings:
Unsupervised HMM captures 94.3% of recession days without labels

YCTI predicts forward volatility
β = 0.0165 | t = 2.14 | p = 0.032
Newey-West corrected | n = 6,268 days

Strategy	Sharpe Ratio	Max Drawdown	Calmar Ratio
Buy & Hold SPY	0.451	-59.6%	0.145
Vol-Targeting	0.596	-27.6%	0.240
YCTI Overlay	0.644	-26.3%	0.253

Cross-Market Validation (no retraining)

Germany: 74% recession capture (5 recessions)
UK: 82% recession capture (4 recessions)
Canada: 100% recession capture (2 recessions) 15 total independent recession episodes

*YCTI is a macro risk conditioning tool, not a standalone forecast model.